Faculty of Computer Science and Mathematics
Universiti Malaysia Terengganu

LAB REPORT

CSE3023 Web-Based Application Development

Name	Wai Pei Yuen
Matric No.	S74991
Semester	2025/26-2
Lab Report	4
Demonstrator	Ts. Dr. Wan Nural Jawahir Hj Wan Yussof

Task 1

Reflection

1. What have you learnt from this exercise?

I learned how to use JSP Scriptlets to perform arithmetic operations and handle condition if-else logic and JSP Expressions to output calculated results.

2. Explain three (3) types of JSP scripting elements.

1) JSP Scriptlets <% … %>

-used to contain Java Language statements, variable declarations or expressions

2) JSP Expression <%= … =>

-used to generate dynamic web content.

3) JSP Declaration <%! … %>

-used to define variables, constants, and methods for reusable logic within JSP pages.

Task 2

Reflection

1. What have you learnt from this exercise?

I learned how to JSP scripting to perform a currency conversion.

Task 3

Reflection

1. What have you learnt from this exercise?

I learned how to use the <jsp:include> to pull in content from another JSP page at runtime.

2. List TWO (2) other JSP Standard Action tags.

<jsp:forward> and <jsp:useBean>.

Task 4

Reflection

1. What have you learnt from this exercise?

I learned how to use <jsp:forward> to redirect a request from one JSP page to another JSP page on the server side.

2. List TWO (2) more JSP Standard Action Tag.

<jsp:setProperty> and <jsp:getProperty>.

Task 5

Reflection

1. What have you learnt from this exercise?

I learned how to implement complex business logic using JSP Scriptlets to calculate insurance premiums, NCD deduction and SST charges.

2. List all Java features you used in Java Scriptlet.

- Variable declaration and initialization: double price = 0;
- Conditional logic: if-else to determine insurance rates.
- Exception handling: try-catch blocks to manage non-numeric input.
- Arithmetic operators: *, -, + to calculate totals and taxes.

Exercise:

Source Code:

header.jsp

```
<%--
    Document   : header.jsp
    Created on : Apr 22, 2026, 10:47:13 PM
    Author     : wpy92
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>Header Page</title>
        <link rel="stylesheet" href="style.css"/>
    </head>
    <body>
        <div class="container">
            <h1>Healthy Lifestyle Awareness Program</h1>
            <nav style="margin-bottom: 20px;">
                <a href="bmiCalculator.jsp">BMI Calculator</a> |
                <a href="resultBMI.jsp">Result</a> |
                <a href="healthInfo.jsp">Health Information</a>
            </nav>
        </div>
    </body>
</html>
```

footer.jsp

```
<%--
    Document   : footer.jsp
    Created on : Apr 22, 2026, 10:51:22 PM
    Author     : wpy92
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>Footer Page</title>
        <link rel="stylesheet" href="style.css"/>
    </head>
    <body>
        <div class="container">
            <hr>
            <p style="text-align: center;">&copy; 2026 FSKM UMT</p>
        </div>
    </body>
</html>
```

bmiCalculator.jsp

```jsp
<%--
    Document   : bmiCalculator.jsp
    Created on : Apr 22, 2026, 10:55:47 PM
    Author     : wpy92
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>BMI Calculator</title>
        <link rel="stylesheet" href="style.css"/>
    </head>
    <body>
        <%@include file="header.jsp" %>

        <div class="container">
            <div class="card">
                <h2 class="form-title">BMI calculator</h2>
                <form action="processBMI.jsp" method="POST">
                    <div class="form-group">
                        <label for="weight">Weight (kg)</label>
                        <input type="number" id="weight" name="weight" step="0.01" required>
                    </div>

                    <div class="form-group">
                        <label for="height">Height (m)</label>
                        <input type="number" id="height" name="height" step="0.01" required>
                    </div>

                    <div class="button-group">
                        <button type="submit" class="btn btn-submit">Submit</button>
                        <button type="reset" class="btn btn-cancel">Cancel</button>
                    </div>
                </form>
            </div>
        </div>

        <%@include file="footer.jsp" %>
    </body>
</html>
```

processBMI.jsp

```
<%--
    Document   : processBMI
    Created on : Apr 22, 2026, 11:04:53 PM
    Author     : wpy92
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>Process BMI</title>
        <link rel="stylesheet" href="style.css"/>
    </head>
    <body>
        <%
            // Retrieve and Convert values
            String weightString = request.getParameter("weight");
            String heightString = request.getParameter("height");

            double bmi = 0;
            String category= "";
```

```
        try {
            if (weightString !=null && heightString !=null) {
                double weight = Double.parseDouble(weightString);
                double height = Double.parseDouble(heightString);

                //Check for height must not be zero
                if (height > 0) {
                    // Calculate BMI
                    bmi = weight / (height * height);

                    // Determine Category
                    if (bmi < 18.5) {
                        category = "Underweight";
                    }

                    else if (bmi >= 18.5 && bmi <= 25) {
                        category = "Normal";
                    }

                    else {
                        category = "Overweight";
                    }
                }

                else {
                    category = "Invalid: Height cannot be zero";
                }
            }
        }

        catch (NumberFormatException e) {
            // Validation: Input must be numeric
            category = "Invalid: Please enter numeric values only";
        }
    %>

    <jsp:forward page="resultBMI.jsp">
        <jsp:param name="bmiValue" value="<%= String.valueOf(bmi) %>" />
        <jsp:param name="bmiCategory" value="<%= category %>" />
    </jsp:forward>
  </body>
</html>
```

resultBMI.jsp

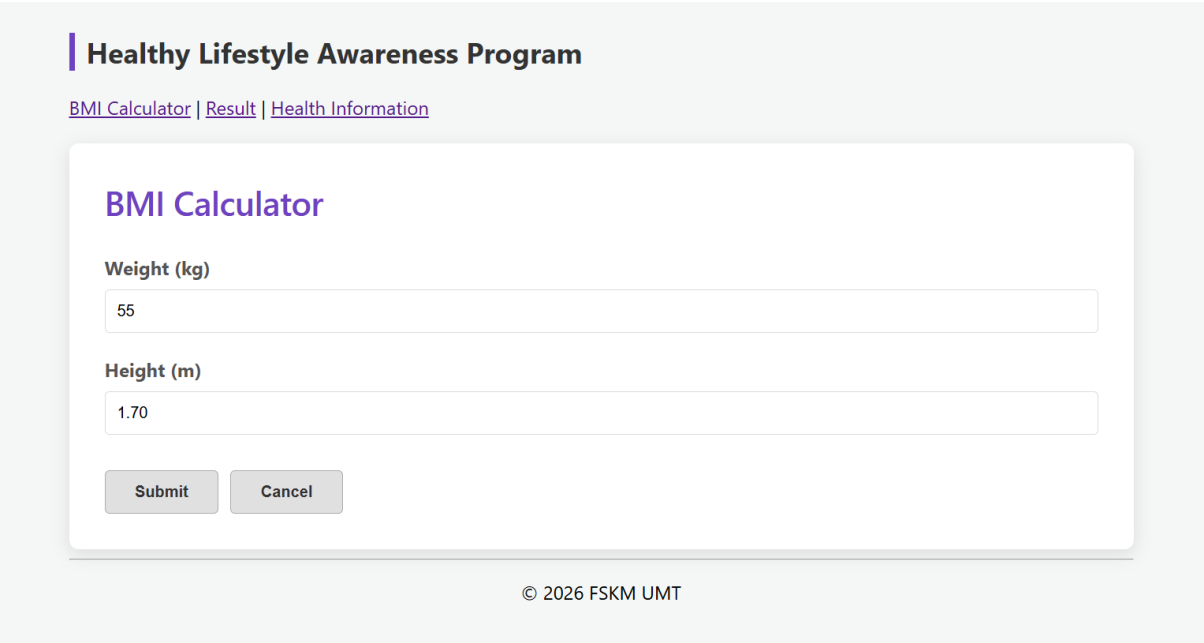
```
<%--
    Document   : resultBMI
    Created on : Apr 22, 2026, 11:18:44 PM
    Author     : wpy92
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>BMI Result</title>
        <link rel="stylesheet" href="style.css"/>
    </head>
    <body>
        <%@include file="header.jsp" %>

        <div class="container">
            <div class="card">
                <h2 class="form-title">Your BMI Result</h2>

                <% if (request.getParameter("bmiValue") == null) { %>
                <p>Please calculate your BMI first using the calculator.</p>
                <% } else { %>

                    <div class="result-group">
                        <label>Calculated BMI:</label>
                        <p><%= String.format("%.2f", Double.parseDouble(request.getParameter("bmiValue"))) %></p>
                    </div>

                    <div class="result-group">
                        <label>Health Category:</label>
                        <p><strong><%= request.getParameter("bmiCategory") %></strong></p>
                    </div>

                    <div class="button-group">
                        <a href="bmiCalculator.jsp" class="btn btn-back"> Calculate Again</a>
                    </div>
                <% } %>
            </div>
        </div>

        <%@include file="footer.jsp" %>

    </body>
</html>
```

healthInfo.jsp

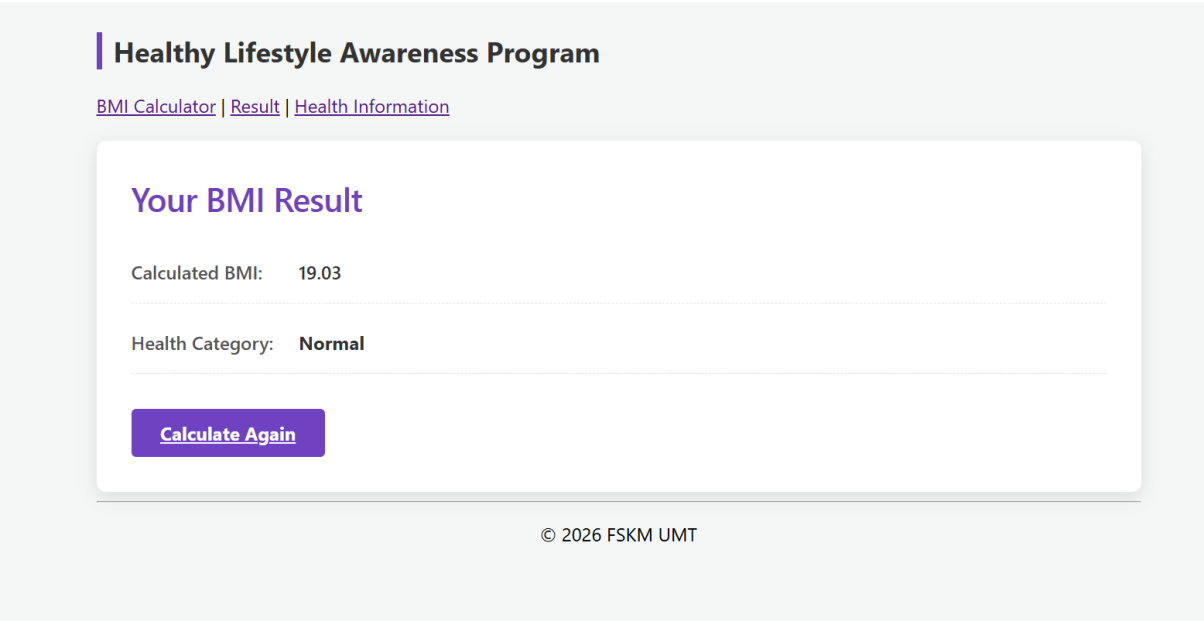
```jsp
<%--
    Document   : healthInfo
    Created on : Apr 22, 2026, 11:26:18 PM
    Author     : wpy92
--%>

<%@page import="java.util.ArrayList"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>Health Info</title>
        <link rel="stylesheet" href="style.css"/>
    </head>
    <body>
        <%@include file="header.jsp" %>

        <div class="container">
            <div class="card">
                <h2 class="form-title">BMI calculator</h2>

                <%
                    ArrayList<String> categories = new ArrayList<>();
                    categories.add("Underweight: BMI < 18.5");
                    categories.add("Normal: 18.5 <= BMI <= 25");
                    categories.add("Overweight: BMI > 25");
                %>

                <table border="1" style="width:100%; border-collapse: collapse; padding: 10px;">

                    <tr style="background-color: #6f42c1; color: white;">
                        <th style="padding: 10px">Classification</th>
                    </tr>

                    <% for(String category: categories) { %>

                    <tr>
                        <td style="padding: 10px"><%= category %></td>
                    </tr>
                    <% } %>
                </table>
            </div>
        </div>

        <%@include file="footer.jsp" %>
    </body>
</html>
```

Output:

BMI Calculator



Result



Health Information

Healthy Lifestyle Awareness Program

BMI Calculator | Result | Health Information

BMI Calculator

Classification
Underweight: BMI < 18.5
Normal: 18.5 <= BMI <= 25
Overweight: BMI > 25

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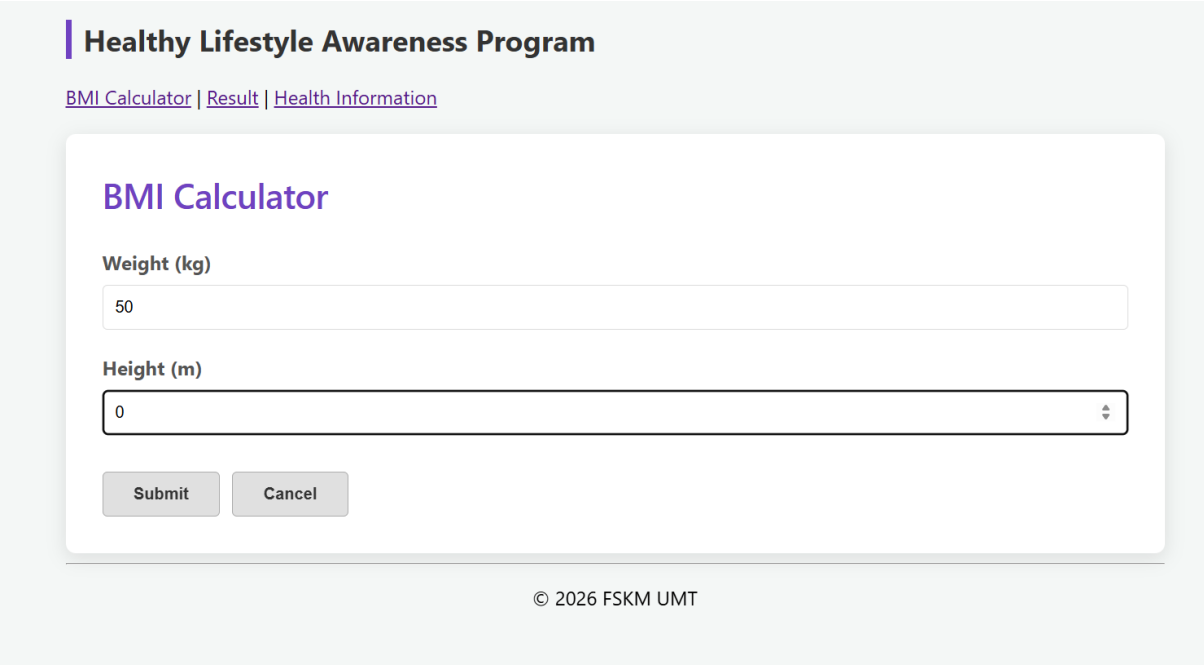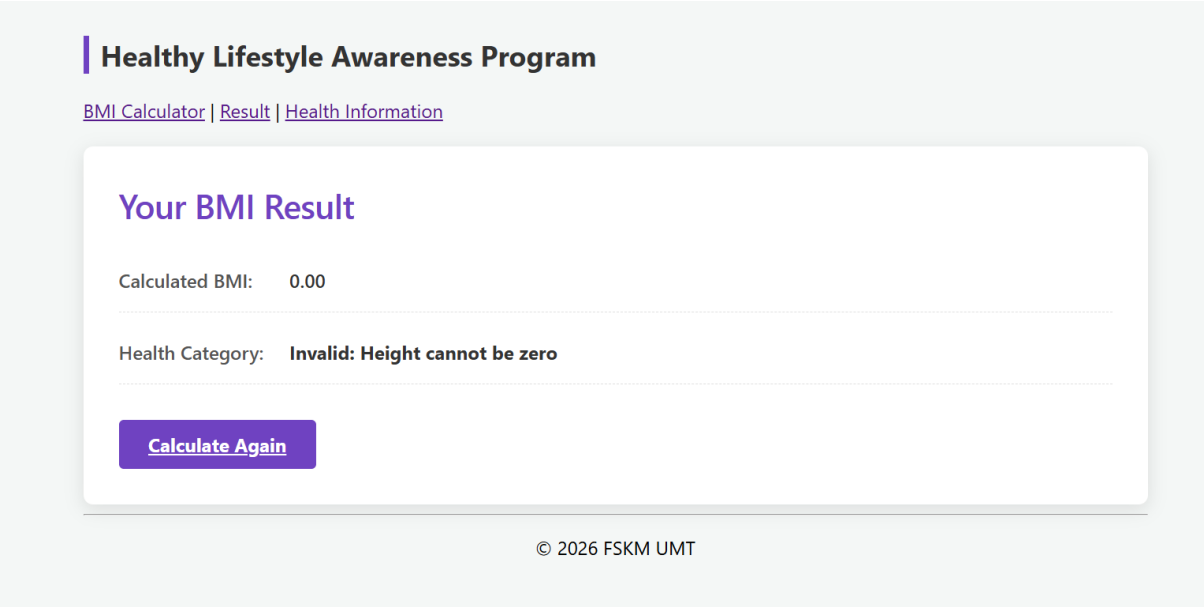
Validation: Input must be numeric and height cannot be zero.